MIL-I-52701A(ME)
19 September 1978
SUPERSEDING
MIL-I-52701
1 February 1971

MILITARY SPECIFICATION

INDICATOR SOLUTION, pH DETERMINATION

(RANGE 3.0 to 11.0)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an indicator solution for the determination of pH values of water in the range 3.0 to 11.0.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-300 - Chemicals, Liquid: Packaging and Packing of.

FSC 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:30Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-D-537 - Disks, Color Standards.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Research Paper 2619 - pH Values of the Clark and Lubs Buffer Solutions at 25° C.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The indicator solution shall produce a color for specific pH values of water that visually match color standards of a master color disk. The solution shall be suitable for use in determining pH values of water from 3.0 to 11.0 in increments of 1 pH.

3.1.1 Standard sample. A standard sample of indicator solution will be supplied by the Government on request by a potential bidder. The standard sample will be furnished only to show the potential bidder the color (obtained when 0.50 ml of standard sample is added to 15 ml of each buffer solution prepared in accordance with 4.4) that the indicator solution he proposes to furnish must produce (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:30Z
Check the source to verify that this is the current version before use.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.3):

Item No.	Description	Quantity
1.	Disk, master color standard, medium values, pH 3.0 to 11.0	1
2.	Disk, master color standard, low values, pH 3.0 to 11.0	1
3.	Disk, master color standard, high values, pH 3.0 to 11.0	1
4.	Disk evaluator.	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:30Z
Check the source to verify that this is the current version before use.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. Failure of the test shall be cause for rejection of the lot represented.

4.4 Inspection procedure.

4.4.1 Examination. The solution shall be examined as specified herein for the following defect:

Major

101. Suspended matter or other foreign material in the solution.

4.4.2 Test. Buffer solutions of pH 3.0 to 11.0 inclusive in increments of 1 pH shall be prepared in accordance with NBS Research Paper 2619. Add 0.50 ml of indicator solution to 15 ml of each buffer solution and compare each color produced with Government-loaned low, medium, and high master color standard disks. Failure of the color produced by the indicator solution to match within the range of color of the low and high value master color standard disks shall constitute failure of this test.

4.5 Inspection of packaging. The preservation, packing, and marking of the indicator solution for Level A or Level B shall be examined for conformance to the quality assurance provisions of PPP-C-300. The preservation, packing, and marking for Commercial shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The indicator solution, in quantities of 4 ounces or 16 ounces, as specified (see 6.2), shall be preserved in accordance with PPP-C-300, using Type III, Class I containers.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:30Z
Check the source to verify that this is the currentversion before use.

5.1.2 Commercial. The indicator solution, in quantities of 4 ounces or 16 ounces, as specified (see 6.2), shall be preserved in accordance with MIL-STD-1188, using plastic containers.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A or Level B. Packing shall be in accordance with PPP-C-300 for the level specified.

5.2.2 Commercial. Packing shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The indicator solution covered by this specification is intended for use in determining the pH of water by comparison with color disks conforming to MIL-D-537, Type I, Class B.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When Government-loaned property is not required (see 3.3).
(c) Whether unit quantities shall be 4 ounces or 16 ounces for Level A or Commercial (see 5.1 and 5.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Government-loaned property. The contracting officer should arrange to loan property specified in 3.3, when applicable (see 6.2).

6.4 Standard sample. The contracting officer should arrange to furnish potential bidders a bottle of indicator solution, pH wide range, NSN 6810-00-087-2340.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - MD
DLA - GS

Project 6810-AA89

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-I-52701A(ME) Indicator Solution, pH Determination (Range 3.0 to 11.0)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:30Z
Check the source to verify that this is the current version before use.